Discrete Uniform Distribution

A discrete r.v variable X is said to follow a uniform distribution if $prob(x) = \frac{1}{n}, x = 1,2,3...$

mgf is $M_x(t) = E(e^{tx}) = \sum_x e^{tx} prob(x) = \frac{e^t(1-e^{tn})}{n(1-e^t)}$ mean $\mu_1' = E(X) = \sum_x xprob(x) = \frac{n+1}{2}$

variance is $\sigma^2 = E(X^2) - \{E(X)\}^2 = \frac{n^2-1}{12}$

Example: Rolling die follow discrete uniform distribution with $p(x) = \frac{1}{6}$

Bernoulli Distribution

A discrete r,v X which takes two values 0 or 1 with probability q and p respectively, i.e., prob(x=0)=q and prob(x=1)=p, such that p+q=1 is called a Bernoulli variate and said to have Bernoulli distribution.

The r^{th} order moment of X is $E(X^r)=0^r xq+1^r xp = p$, r=1,2,3....

Hence mean of Bernoulli distribution is $\mu_1' = E(X) = \sum_x x \times prob(x) = 1 \times p + 0 \times q = p$

Similarly $\mu_2' = E(X^2) = \sum_x x^2 \times prob(x) = 1 \times p + 0 \times q = p$, then variance of the r.v. X is

$\sigma^2 = E(X^2) - \{E(X)\}^2 = p - p^2 = p(1-p) = pq$

The mgf of X is $M_x(t) = E(e^{tx}) = \sum_x e^{tx} \times prob(x) = e^{t\times 0} \times q + e^{t\times 1} \times p = pe^t + q$

Mean and variance using $M_x(t)$ is $M_x'(t)$ at $t=0$ is $pe^0 = p \times 1 = p$,

$\sigma^2 = E(X^2) - \{E(X)\}^2 = M_x''(t) - \{M_x'(t)\}^2$ at $t=0$ is $\mathrm{p - p^2 = pq}$